

System 1 and System 2 Thinking

What is System 1 and System 2 Thinking?

System 1 and System 2 thinking describes two distinct modes of cognitive processing introduced by Daniel Kahneman in his book *Thinking, Fast and Slow*. **System 1** is fast, automatic, and intuitive, operating with little to no effort. This mode of thinking allows us to make quick decisions and judgments based on patterns and experiences. In contrast, **System 2** is slow, deliberate, and conscious, requiring intentional effort. This type of thinking is used for complex problem-solving and analytical tasks where more thought and consideration are necessary.



The Basic Idea

When commuting to work, you always know which route to take without having to consciously think about it. You automatically walk to the subway station, habitually get off at the same stop, and walk to your office while your mind wanders. It's effortless. However, the subway line is down today.

While your route to the subway station was intuitive, you now find yourself spending some time analyzing alternative routes to work in order to take the quickest one. Are the buses running? Is it too cold outside to walk? How much does a rideshare cost?

Our responses to these two scenarios demonstrate the differences between our instantaneous System 1 thinking and our slower, more deliberate System 2 thinking.

However, even when we think that we are being rational in our decisions, our System 1 beliefs and biases still drive many of our choices. Understanding the interplay of these two systems in our daily lives can help us become more aware of the bias in our decisions—and how we can avoid it.

“The automatic operations of System 1 generate surprisingly complex patterns of ideas, but only the slower System 2 can construct thoughts in an orderly series of steps.

—Daniel Kahneman in *Thinking, Fast and Slow*

System 1 Thinking: Our brains' fast, automatic, unconscious, and emotional response to situations and stimuli. This can be in the form of absentmindedly reading text on a billboard, knowing how to tie your shoelaces without a second thought, or instinctively hopping over a puddle on the sidewalk.

System 2 Thinking: The slow, effortful, and logical mode in which our brains operate when solving more complicated problems. For example, System 2 thinking is used when looking for a friend in a crowd, parking your vehicle in a tight space, or determining the quality-to-value ratio of your take-out lunch.

Automatic Thinking: An unconscious and instinctive process of human thinking. This term can be used interchangeably with System 1 thinking.

Reasoning: Consciously using existing information to logically make a decision or reach a conclusion, a key feature of System 2 thinking.

Dual Process Model: A theory in psychology that distinguishes two thought processes in humans by describing them as unconscious and conscious, respectively.

Behavioral Economics: A field that examines how psychological factors influence economic decision-making, often through the interaction of System 1 and 2 thinking. It highlights how cognitive biases and emotions can lead to deviations from rational behavior in financial choices.

For many centuries, philosophers and psychologists have been able to differentiate instinctive thinking and conscious reasoning, starting as early as the 17th century with Descartes' mind-body dualism.

William James, an American psychologist, was at the root of this idea in the late 19th century. In his book, *Principles of Psychology*, James believed that associative and true reasoning formed the two ways of thinking.^{1,2} Associative knowledge was derived only from past experiences, as opposed to true reasoning being used in new, unfamiliar scenarios that an individual is unfamiliar with. James's ideas laid the groundwork for System 1 and System 2 thinking.

In 1975, psychologists Michael Posner and Charles Snyder developed the dual-process model of the human mind in their book, *Attention and Cognitive Control*. The dual-process model was a more nuanced version of James' ideas, distinguishing the two ways of thinking by describing them as **automatic** and controlled, respectively.³

As the theory developed, automatic processes were characterized by four conditions:

1. They are elicited unintentionally;
2. They require only a small amount of cognitive resources;
3. They cannot be stopped voluntarily; and
4. They happen unconsciously.

Likewise, controlled processes were characterized by four conditions:

1. They are elicited intentionally;
2. They require a considerable amount of cognitive resources;
3. They can be stopped voluntarily; and
4. They happen consciously.

However, in 1992, John Bargh challenged these rigid characteristics and suggested that it was virtually impossible for any process to satisfy all four.⁴

Fast forward to 2011, and Daniel Kahneman published his bestselling book, *Thinking, Fast and Slow*, popularizing the distinction between automatic and conscious thought processes.⁵ In this book, Kahneman incorporated the terms System 1 and System 2 to describe the two processes, first coined by psychologists Keith Stanovich and Richard West in 2000.⁶ His work also illuminated the pervasive impact of cognitive biases, showing how the interplay between these systems often leads to systematic errors in judgment. Kahneman's contributions have fundamentally shaped our understanding of human decision-making, blending psychological insights with economic theory.

“We are all familiar with the experience of trying not to stare at the oddly dressed couple at the neighboring table in a restaurant. We also know what it is like to force our attention on a boring book, when we constantly find ourselves returning to the point at which the reading lost its meaning... And every human being has had the experience of not telling someone to go to hell. One of the tasks of System 2 is to overcome the impulses of System 1. In other words, System 2 is in charge of self-control.”

Daniel Kahneman¹²

People

Daniel Kahneman

A renowned psychologist in the field of **behavioral economics** who was influential in topics such as judgement and decision-making. Kahneman's 2011 book, *Thinking, Fast and Slow*, popularized the concepts of System 1 & System 2.

William James

An American psychologist, philosopher, and historian who is credited with laying the initial groundwork for two different types of thinking in the late 19th century. His work would go on to influence formal literature on the dual process model in the late 20th century. At Harvard University, James was one of the very first educators to offer a psychology course in the United States.⁷

Michael Posner

An American psychologist who, along with Charles Snyder, was one of the first to formally introduce the dual process model. Posner and Snyder's book, *Attention and Cognitive Control*, described the two forms of thinking as automatic and controlled, respectively.

Can System 1 and System 2 thinking happen at the same time?

Yes! In fact, this is most often the case. Rather than just relying on the intuitive and fast thinking of System 1 or the intentional and slow thinking of System 2, our brains usually use a combination of both to solve a problem or complete a task. Both modes tend to complement each other, filling in the gaps that the other misses. Let's take the example of baking a new recipe. Your basic cooking skills—such as whisking eggs or measuring flour—most likely fall into System 1 thinking, allowing you to follow the instructions quickly and smoothly without putting in too much forethought. This saves time and mental energy for you to devote deliberate System 2 thinking to new skills—such as piping frosting or caramelizing sugar. Together, a balanced combination of System 1 and System 2 can not only help you whip up some delicious baked goods but also complete more tasks with focus *and* ease.

What do heuristics have to do with System 1 or System 2 thinking?

Heuristics are mental shortcuts that help us surpass analytical thinking and jump right to immediate judgments or automatic responses. As you may have guessed, these “generalizations” or “rules-of-thumb” lessen our cognitive load—meaning they usually fall under the category of System 1 thinking. (However, this is not to say that System 2 isn't susceptible, as you can read more about in our “Consequences” section.) Most common heuristics can be a powerful tool for getting things done fast and saving time for other, more demanding tasks... but, unfortunately, these don't always lead to the most accurate answers. This why we must be sure to constantly be identifying, addressing, and refining our shortcuts with logical reasoning to make sure that they are supporting our cognitive processes, rather than depleting them.

Is System 1 or System 2 better?

Neither System 1 or System 2 is fully “better”—however, one may be more helpful than the other, depending on what we're trying to accomplish. For instance, System 1 is usually better for repetitive, everyday tasks that are so habitual that they require few cognitive abilities. After all, imagine having to think through how to complete household chores like making the bed or washing the dishes every single time that we do them. That would be exhausting! Meanwhile, System 2 is typically better reserved for a novel, effortful activity—especially when we are making decisions that have greater implications. These consequential choices in our education, careers, and personal lives are where we need System 2 thinking to steer us in the right direction, rather than let System 1 take the wheel.

Can an activity switch from System 2 to System 1 thinking?

Of course! In fact, the switch from System 2 to System 1 thinking plays a vital role in **habit formation**—where mental activities become quicker and easier the more that we repeat them. Let's go back to our chore example. Although making the bed may seem easy now, imagine yourself as a young child trying to do this for the first time. Rather than performing the task automatically, you most likely had to engage in System 2 thinking to review each step of the process. (First, strip the bed. Then, tuck in the sheets. Then, fluff the pillows...) However, as you got older, and (hopefully) kept making your bed more and more, there was less need for such intentional thought, allowing you to downshift to System 1.

Marketing

The concepts of System 1 and System 2 have become highly influential in the world of marketing. In a world where consumers have more options than ever, brands often rely on the automatic, feelings-driven processes of System 1 to sell their products. Advertising seeks not just to communicate information about a product, but also to establish certain emotional associations around it that will stick in customers' heads and drive them to purchase it without extra thought.



The power of System 1 thinking means that overhauled and refreshed marketing campaigns may not be as effective as initially thought. As competition continues to grow fiercely in the field of marketing, many brands are attempting bold, radically new campaigns. However, overhauled campaigns may get rid of the valuable, distinctive features that shape consumers' automatic perception of the brand's image.⁸ By leveraging a brand's distinctive image to increase its resonance in consumers' System 1 thinking, a greater return on investment can be created in the short and long term.⁸

That doesn't mean that System 2 doesn't play an important role in consumer decisions. For expensive purchases, consumers tend to make decisions based on System 1 beliefs, in addition to a more careful and rational thought process driven by System 2.⁸ Brands can use their knowledge of System 2 to provide a powerful justification, reinforcing consumers' System 1 beliefs with details, facts, or statistics.⁸

Financial Planning

Governments can also take advantage of System thinking to develop effective behavioral interventions. Recognizing System 1 thinking's automatic preference for the [default](#) has led to the development of effective interventions, addressing issues such as insufficient retirement savings.⁹

In the United States, behavioral economists recognized that even when workers received a raise, few would actually take action to increase their savings rate. They concluded that the lack of action was a sign of an overreliance on System 1 thinking.

In this case, the default option kept the savings rate the same, unless a worker took action to increase it. To tackle the problem, behavioral economists designed an intervention that automatically increased a worker's savings rate whenever they received a raise. The automatic increase was able to take advantage of workers' System 1 thinking to increase savings rates in the US.⁹

The concepts of System 1 and System 2 thinking have become common in mainstream thinking. The transition from academia to popular culture has resulted in the original theory losing some of its nuance and depth, replaced by simplifications of human thought processes. There are three common misconceptions that have emerged in popular culture.⁵

Misconception 1: Systems 1 and 2 are distinctive brain structures

First is the idea that System 1 and System 2 thinking literally represent two separate parts of the human mind—similar to the pop-psych concept of “left brain” and “right brain” thinking. This brain structure is false, and Kahneman even says that “there is no part of the brain that either of the systems would call home.”¹⁰

**Misconception 2: First System 1, then System 2**

Second is the idea that System 1 thinking occurs first, followed by System 2 thinking if necessary. Kahneman explains that the dual-system approach combines both forms of reasoning as almost all processes are a mix of both systems. Though difficult scenarios may rely more on System 2, both systems work together. Emotions from our unconscious System 1 processes influence and complement our logical System 2 thinking, and our brain integrates the two to enable us to make purposeful decisions.⁵

Misconception 3: System 1 creates bias, and System 2 fixes it

Finally, popular culture tends to incorrectly label System 1 as the source of bias, and System 2 as the logical correction to said biases. In fact, both systems are susceptible to biases and mistakes, such as [confirmation bias](#).⁵ For example, we may notice information when it supports our existing System 1 beliefs, in addition to using System 2 to analyze new information in order to justify our existing beliefs as a result of the confirmation bias.⁵

The Invisible Gorilla

Would you spot a gorilla if it walked right across the screen? If you think the answer is an obvious “yes,” think again.

In 1999, psychologists Christopher Chabris and Daniel Simons conducted a now-famous experiment to illustrate the phenomenon of inattention blindness.¹³ Participants were asked to watch a video in which two teams, one dressed in white and the other in black, passed basketballs back and forth. The task was simple: count the number of passes made by the team in white. Focused on this task, many viewers failed to notice something unusual—a person in a gorilla suit walking into the frame, beating their chest, and then walking off-screen. When asked afterward, most participants were shocked to learn that they had missed such an obvious detail.



This experiment highlights the limitations of our attention and how easily we can miss significant information when our cognitive resources are focused elsewhere. The connection to System 1 and System 2 thinking is clear: the task of counting passes engaged System 2, which is deliberate, effortful, and analytical. However, this intense focus on the counting task left participants blind to the unexpected gorilla, a task that would typically fall under the automatic, intuitive processing of System 1. The experiment demonstrates how System 2 can sometimes dominate our attention so completely that System 1 fails to register even the most conspicuous elements in our environment.

The invisible gorilla serves as a powerful metaphor for ethical inattention blindness in decision-making.¹⁴ Just as participants missed the gorilla while focusing on counting passes, individuals and organizations can overlook obvious ethical issues when their attention is consumed by other tasks or goals. The experiment underscores the importance of balancing System 1 and System 2 thinking, ensuring that while we engage in detailed analysis, we do not lose sight of broader, intuitive signals that might alert us to critical issues.

“...you can also feel a surge of conscious attention whenever you are surprised. System 2 is activated when an event is detected that violates the model of the world that System 1 maintains. In that world, lamps do not jump, cats do not bark, and gorillas do not cross basketball courts.

Daniel Kahneman¹⁴

M&M Advertising

In 1995, the popularity of M&M’s, the multi-colored chocolate candy, was decreasing. BBDO, an advertising agency, was recruited in an attempt to revitalize the brand. Then-creative director, Susan Credle, had a small budget to work with compared to other iconic brands, like Pepsi or Coke. However, Credle’s approach was highly successful: she made each color of M&M candy into a character—a ‘spokescandy’.¹¹ BBDO introduced Red (the sarcastic one), Yellow (the happy one), Blue (the cool one), and Green (the flirty one).



This move resulted in the creation of M&M retail stores and multiple M&M line extensions.¹¹ The characters became so popular that, in an attempt to prevent consumers from losing interest, BBDO experimented with occasionally removing

remain easily identifiable.

Automatic Thinking

By developing memorable characters, BRDO was able to successfully ingrain The Decision Lab takes a closer look at automatic thinking by considering its M&M into consumers' System 1 thinking. This was achieved on a sustainable history, in addition to the consequences and controversies, it is associated with mass scale by creating distinctive brand assets. This not only deepens M&M's resonance in consumers' System 1 thinking it also creates more return on investment in the short and long run.

Although aging is inevitable, financial fraud in old age isn't. Elderly individuals in the US alone lose an estimated \$3 billion a year to financial scams. System 1 thinking can play a part in this, and research by The Decision Lab offers insights into how this reality can be avoided.

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